

Augmenting Vision-Language Models with Spatially-Aware Dual Encoders for Enhanced Visual Grounding

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Abstract

Contemporary Vision-Language Models (VLMs) demonstrate robust global semantic comprehension but frequently struggle with fine-grained spatial reasoning tasks, such as object counting and depth estimation. This deficiency primarily stems from the inductive biases of standard CLIP-variant vision encoders, whose reliance on natural language supervision inherently prioritizes instance-level semantic alignment at the expense of localized geometric fidelity. To address this representational bottleneck, we propose a dual-encoder architecture that explicitly integrates the spatially dense, self-supervised features of DINOv2 into a baseline VLM (Qwen2.5-VL) via a straightforward token-level fusion mechanism. We rigorously evaluate the augmented framework across spatially demanding benchmarks, including CV-Bench, CLEVR, TallyQA, and VSR. Our quantitative results demonstrate substantial performance gains, while qualitative analyses using Principal Component Analysis (PCA) and patch-level cosine similarity confirm that the DINOv2 integration yields sharply localized semantic boundaries and mitigates spatial feature leakage. Furthermore, an ablation study on the fusion scaling factor α , examined via layer-wise spatial attention entropy, reveals a fundamental trade-off: integrating geometric priors significantly enhances visual grounding, yet excessive spatial structure disrupts the language model’s semantic alignment. Ultimately, this work validates the necessity of explicit spatially-aware representation streams and provides a robust blueprint for developing geometrically grounded VLMs without the need for complex, parameterized fusion modules.

1 Introduction

While contemporary Vision-Language Models (VLMs) demonstrate robust global semantic comprehension, they frequently exhibit suboptimal performance on fine-grained spatial reasoning tasks, such as object counting, depth estimation, and resolving topological relations [1]. This deficiency can be fundamentally attributed to the inductive biases inherent in their visual encoding paradigms. The prevailing reliance on CLIP-variant vision encoders—adopted for their effective cross-modal alignment via contrastive objectives—inherently prioritizes instance-level semantic coherence over localized geometric fidelity. Consequently, these encoders tend to discard dense spatial representations. To address this representational bottleneck, this work investigates the integration of a supplementary, spatially-aware vision encoder to augment structural grounding without necessitating end-to-end retraining.

This investigation proposes a dual-encoder architecture that explicitly fuses spatially dense DINOv2 [2] features into the Qwen2.5-VL [3] baseline at the token level. To quantify the recovery of spatial acuity, the augmented architecture is subjected to rigorous evaluation across spatially demanding datasets: CV-Bench, CLEVR, TallyQA and VSR. [1],[4],[5],[6]. Beyond standard performance metrics, we dissect the integration mechanism through an ablation on the fusion scaling parameter α . This encompasses a granular analysis of task accuracy alongside an examination of the language model’s spatial entropy and PCA-derived feature maps, ultimately demonstrating the framework’s refined visual grounding.

2 Background

2.1 Self-supervised Vision Encoder

Historically, learning visual representations via natural language supervision relied heavily on predictive objectives. In these early frameworks, CNN-derived spatial features were linearly projected and fed into autoregressive decoders, such as LSTMs [7] or Transformers [8], to optimize an exact, token-by-token prediction loss. However, this strict generative formulation exhibited poor scalability and suboptimal performance, as it was heavily bottlenecked by the rigidity of exact-match text generation and highly curated datasets. While subsequent generative representation learning approaches yielded higher-fidelity features, they introduced prohibitive computational overhead [9].

To circumvent the inefficiency of predictive modeling, modern paradigms leverage contrastive learning, which extracts robust, scalable representations at a fraction of the computational cost. Rather than predicting exact tokens, the contrastive objective aligns modalities by pulling the embeddings of matched (positive) image-text pairs closer in a joint latent space while pushing unmatched (negative) pairs apart. This is formalized by:

$$L_i = -\log \frac{e^{z_i z_j / \tau}}{\sum_{k=1, k \neq i}^{2N} e^{z_i z_k / \tau}} \quad (1)$$

where τ is a temperature parameter, and z is a feature embedding vector.

CLIP [10] epitomizes this methodology through a dual-encoder architecture. It processes images via a ResNet or Vision Transformer (ViT) and text via a masked self-attention Transformer. By computing the dot-product similarity between the L2-normalized visual and textual embeddings, CLIP maximizes the similarity of positive pairs across a batch while minimizing it for all negative pairs, enabling highly efficient zero-shot transfer.

While CLIP extracts highly effective image representations, its contrastive objective inherently relies on excessively large batch sizes to supply sufficient negative pairs for optimal performance. Furthermore, standard supervised learning paradigms often force models to learn narrow, task-specific shortcuts to minimize the loss, rather than developing rich, generalized visual features.

To circumvent the heavy reliance on negative sampling, self-supervised bootstrapping paradigms were introduced, most notably DINO [11]. DINO employs a multi-crop strategy that applies varying augmentations to generate two global views and multiple local views of a single image. A student network processes all crops, while a teacher network processes only the global views, thereby explicitly enforcing local-to-global spatial correspondences. Both networks share an identical architecture but maintain separate parameters. The model minimizes the cross-entropy loss between the softmax-normalized probability distributions of the student and teacher.

$$P_s(x)^{(i)} = \frac{\exp(g_{\theta_s}(x)^{(i)} / \tau_s)}{\sum_{k=1}^K \exp(g_{\theta_s}(x)^{(k)} / \tau_s)} \quad (2)$$

$$\min_{\theta_s} H(P_t(x), P_s(x)) \quad (3)$$

Crucially, backpropagation is restricted to the student network, while the teacher’s weights are iteratively updated via an exponential moving average of the student’s weights. A consequence of this self-distillation process is that the self-attention maps corresponding to the class token of the trained student network naturally emerge as highly accurate semantic segmentation masks, demonstrating exceptional dense visual grounding.

2.2 Vision Language Model

While early Vision-Language Models such as Flamingo [11] and BLIP [12] excelled at static image captioning, they critically lacked the capacity for open-ended, user-directed visual instruction following. To bridge this gap, LLaVA [13] introduced a paradigm for visual instruction tuning. Because large-scale, multimodal instruction-following datasets were practically nonexistent, LLaVA bypassed this bottleneck by leveraging a language-only GPT-4 [14] model to synthesize instruction-response pairs. By feeding GPT-4 symbolic image representations—specifically, existing dataset captions and bounding

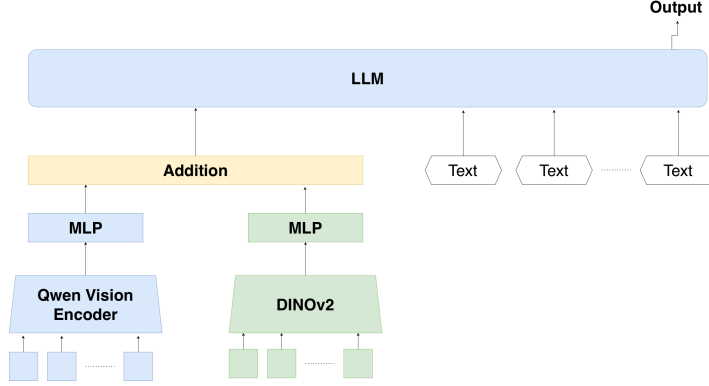


Figure 1: The proposed dual-encoder architecture. The framework processes input images through both the native Qwen vision encoder and the supplementary DINOv2 backbone. The resulting feature maps are independently projected via Multi-Layer Perceptron (MLP) connectors, aggregated via element-wise summation, and prepended alongside text tokens for the large language model (LLM).

box coordinates—the authors generated diverse, high-quality visual QA datasets without requiring the teacher model to process raw images.

The LLaVA training pipeline is executed in two distinct stages. In the first stage, known as feature alignment, both the vision encoder and the large language model (LLM) are frozen; only a linear projection matrix W is trained to map visual features directly into the pre-trained LLM’s word embedding space. In the second stage, visual instruction tuning, the vision encoder remains frozen while both the projection layer and the LLM parameters are fine-tuned end-to-end on the generated multimodal dataset. Throughout both phases, the model optimizes the standard autoregressive next-token prediction objective.

Building upon this foundation, subsequent iterations like LLaVA-1.5 [15] enhanced cross-modal alignment by replacing the rudimentary linear projection with a more expressive Multi-Layer Perceptron (MLP). The design of these vision-language connectors has since seen rapid innovation, with contemporary frameworks introducing advanced modality-bridging mechanisms such as Q-Formers [16] and Spatial Vision Aggregators (SVA) [1]. Collectively, this architectural blueprint and the two-stage visual instruction-tuning paradigm have established the de facto standard for modern Vision-Language Models. This lineage notably includes Qwen-VL, which we adopt as the foundational baseline architecture for our investigation.

3 Approach

We hypothesize that the suboptimal performance of contemporary VLMs on fine-grained visual grounding tasks is fundamentally tied to their reliance on CLIP-based vision encoders, which tend to produce spatially homogeneous feature representations. To address this bottleneck, we investigate whether augmenting the baseline Qwen2.5-VL architecture with a spatially discriminative, self-supervised vision encoder—specifically, DINOv2—can recover the lost spatial acuity.

The proposed dual-encoder architecture shown in Figure 1 processes a single input image in parallel through both the native Qwen vision encoder and the DINOv2 backbone. The distinct feature maps extracted from each encoder are independently projected via an MLP connector to match the language model’s embedding dimension, aggregated via element-wise summation, and finally prepended to the input text tokens for ingestion by the autoregressive LLM. Following the established LLaVA training paradigm, we employ a two-stage training strategy: feature alignment and visual instruction tuning. In the initial alignment phase, the vision encoders and the LLM are strictly frozen; only the weights of the MLP connector are optimized on a randomly sampled 100k subset of the Cambrian-1 1.2M dataset [1] to map the novel visual representations into the LLM’s embedding space. In the subsequent instruction tuning phase, the vision encoders remain frozen while both the MLP connector and the LLM are jointly fine-tuned on a 100k subset of the LLaVA-665k dataset [1].

Crucially, rather than employing complex, parameterized fusion mechanisms, we intentionally adopt

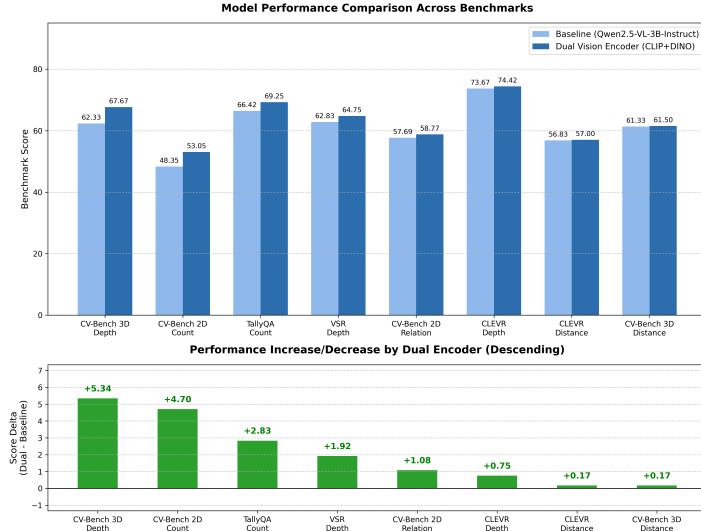


Figure 2: Quantitative performance comparison across visual grounding benchmarks. The top chart illustrates the absolute scores of the baseline Qwen2.5-VL-3B-Instruct model against the proposed dual vision encoder (CLIP+DINO) architecture at fusion scaling factor $\alpha = 1$. The bottom chart highlights the absolute performance delta, demonstrating the most significant gains in depth estimation and object counting tasks.

a straightforward element-wise summation to aggregate the dual visual features. This design choice strictly isolates the architectural variable, ensuring that any observed improvements in visual grounding can be definitively attributed to the supplementary DINOv2 encoder rather than the fusion module itself. Importantly, our gains arise without spatial aggregation or learned routing mechanisms, indicating that improved visual grounding can be achieved purely through representational enrichment rather than architectural complexity. To guarantee a fair comparison, both the baseline and the proposed dual-encoder architectures were trained independently under identical conditions for a single epoch, optimizing the standard autoregressive next-token prediction loss. Computationally, training the 3B-parameter baseline model required 14 hours for alignment and 17 hours for instruction tuning on a node equipped with four 11GB NVIDIA RTX 2080Ti GPUs. Under the same hardware constraints, the dual-encoder architecture required 18 hours for each respective phase.

4 Results

4.1 Quantitative Evaluation on Visual Grounding Benchmarks

Model	CV-Bench				CLEVR		TallyQA	VSR
	3D		2D		Depth	Distance	Count	Depth
	Depth	Distance	Count	Relation				
Baseline (Qwen2.5-VL-3B-Instruct)	62.33	61.33	48.35	57.69	73.67	56.83	66.42	62.83
Dual Vision Encoder (CLIP+DINO)	67.67	61.50	53.05	58.77	74.42	57.00	69.25	64.75
$\alpha = 2$	66.00	61.83	53.43	58.46	73.58	57.17	71.58	63.83
$\alpha = 3$	65.33	60.67	53.43	59.08	72.25	57.50	71.17	64.42
$\alpha = 4$	66.17	61.50	53.43	59.08	68.83	56.25	70.00	63.50
$\alpha = 5$	65.83	58.00	53.55	59.38	66.50	52.50	69.75	63.25

Table 1: Quantitative evaluation across spatial reasoning and visual grounding benchmarks.

The proposed architecture is systematically evaluated across a suite of visual grounding and spatial reasoning benchmarks, specifically CV-Bench, CLEVR, TallyQA, and the VSR evaluation set. Overall, the dual-encoder framework consistently outperforms the baseline across all metrics when the fusion

scaling factor is set to $\alpha = 1$. The most substantial performance gains are observed in the object counting and depth estimation tasks. Regarding counting, DINOv2’s self-supervised features inherently encode both semantic identity and fine-grained spatial localization, enabling the language model to more reliably disambiguate and enumerate individual instances. For depth reasoning, our results corroborate findings from the original DINOv2 literature [2]; just as linear probing on DINOv2 features yields highly accurate monocular depth estimation, our evaluation demonstrates that robust depth cues are already embedded within DINOv2’s patch-level representations, which the augmented VLM successfully leverages.

4.2 Probing the Spatial Awareness of DINOv2 Representations

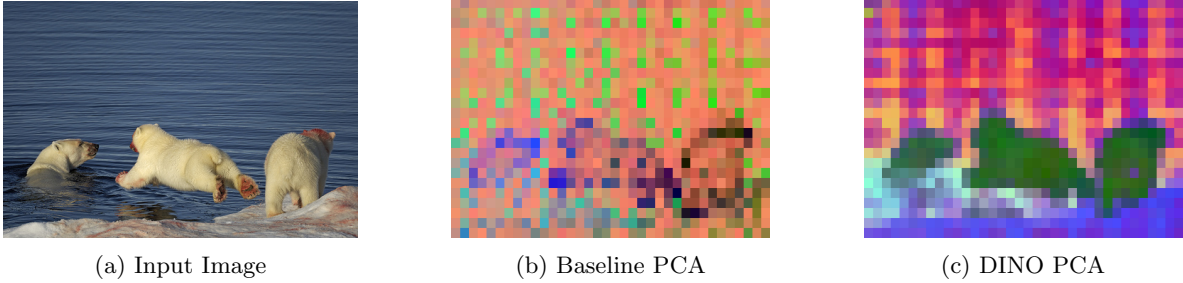


Figure 3: Principal Component Analysis (PCA) visualizations of dense spatial features. By projecting the high-dimensional last hidden states into an RGB subspace, the baseline model (b) exhibits spatially diffused features, whereas the DINOv2-augmented dual-encoder (c) demonstrates sharply localized semantic boundaries and distinct object separation.

To demonstrate the superior object localization capabilities afforded by the integrated DINOv2 features, we perform Principal Component Analysis (PCA) on the last hidden state representations of both the dual-encoder architecture and the baseline model. The high-dimensional feature vectors are projected into a three-dimensional subspace, where each principal component is mapped to the RGB color channels to visualize semantic and spatial variance. The resulting patch-level representations are reconstructed into a spatial grid and upsampled to the original image dimensions using a nearest-neighbor interpolation algorithm to preserve localized feature boundaries.

As illustrated in Figure 3, the PCA visualizations for the dual-encoder model exhibit a distinct and clear separation between disparate objects and background elements, whereas the baseline features appear more spatially diffused. This qualitative evidence confirms that the DINOv2 pathway successfully injects dense semantic identity and precise coordinate information into the language model’s latent space, facilitating robust visual grounding.

To further quantify the spatial coherence of the learned representations, we evaluate the pairwise cosine similarity between a designated reference patch and the broader grid of spatial tokens. Extracting the last layer feature embeddings, we compute the cosine similarity and project the resulting affinity maps back onto the original image. To accurately reflect the discrete boundaries of the tokens without introducing smoothing artifacts, the spatial resolution is restored utilizing zero-order hold scaling. As depicted in Figure 4, the proposed dual-encoder architecture demonstrates highly concentrated affinity, with elevated similarity scores strictly confined to the target instance. In contrast, the baseline model exhibits considerable feature leakage into surrounding regions. This pronounced intra-object coherence provides compelling evidence that the supplementary DINOv2 stream successfully injects discriminative geometric priors, thereby reinforcing the model’s capacity for complex spatial reasoning.

Overall, the proposed dual-encoder framework, augmented with DINOv2 representations, significantly outperforms the baseline across multiple visual grounding and spatial reasoning benchmarks, yielding particularly strong improvements in object counting and depth estimation. To elucidate these performance gains, qualitative analyses using Principal Component Analysis (PCA) and patch-level cosine similarity were conducted on the models’ spatial representations. The resulting visualizations demonstrate that the dual-encoder achieves sharply localized semantic boundaries and highly concentrated intra-object feature affinity. In contrast to the baseline, which suffers from spatial diffusion and

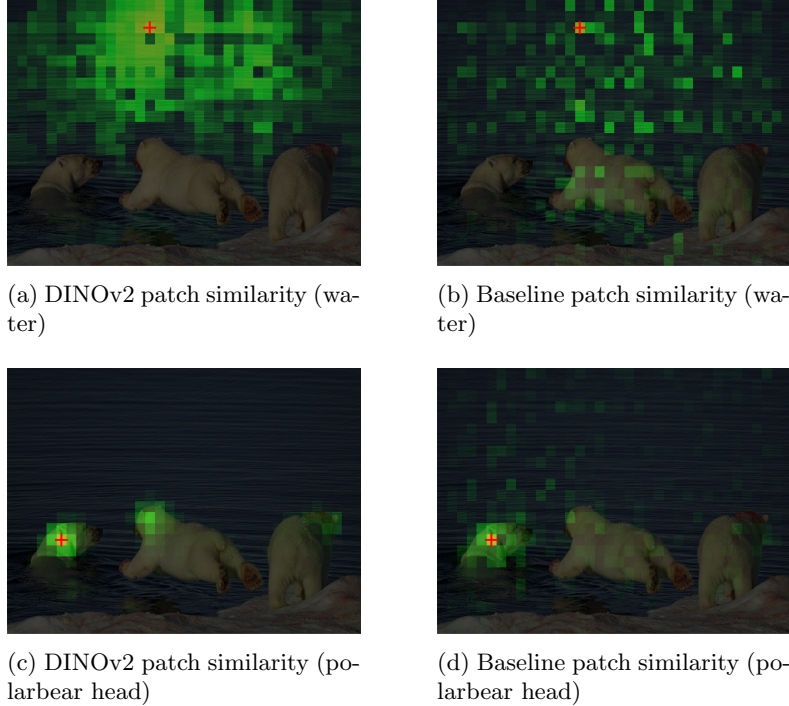


Figure 4: Patch-level cosine similarity affinity maps. Visualizations compare the baseline and DINOv2-augmented architectures using designated reference patches for the background (water) and a specific instance (polar bear head). The dual-encoder exhibits highly concentrated intra-object feature affinity, effectively mitigating the spatial feature leakage observed in the baseline.

feature leakage into surrounding regions, the integration of DINOv2 successfully injects precise geometric priors and dense coordinate information, thereby fundamentally enhancing the model’s overall spatial awareness and visual grounding capabilities.

4.3 Ablation Study on the Fusion Scaling Factor

To further investigate the internal attention dynamics, we conduct an ablation study on the fusion scaling factor α to observe its impact on multimodal integration. As shown in 1 and 5, we can see increasing the fusion ratio peaks the performance at certain α , and start to decrease.

To elucidate the internal attention dynamics of the language model, we analyze its spatial entropy across all layers. For every attention head within the architecture, we extract the self-attention weights. Because the softmax operation inherently constrains these weights to sum to one, they form a valid discrete probability distribution. We compute the Shannon entropy of this distribution for each spatial query token and subsequently average these per-query values to derive a representative spatial entropy metric for each respective head. These layer-wise and head-wise entropy metrics are visualized as a heatmap in Figure 6. The visualization demonstrates that as the fusion scaling factor α increases, the spatial entropy within the deeper layers of the network progressively decreases. This reduction signifies that the model’s attention becomes highly concentrated on specific, localized regions rather than being uniformly distributed across the visual space. Given that deeper layers typically encode high-level semantic abstractions [17], this sharply focused attention indicates a substantially enhanced visual grounding capability.

Conversely, when the fusion scaling factor α exceeds its optimal threshold, the spatial entropy within the deeper layers exhibits a significant increase, correlating with a notable degradation in overall performance. Initially, elevating α successfully injects robust spatial priors into the fused token representations, which sharpens the spatial attention of the language model and enhances visual grounding. However, at excessively high values of α , the supplementary DINOv2 representations overwhelmingly dominate the language-supervised features inherent to the Qwen backbone. Under these conditions, the model retains the capacity for precise object localization but suffers a critical loss

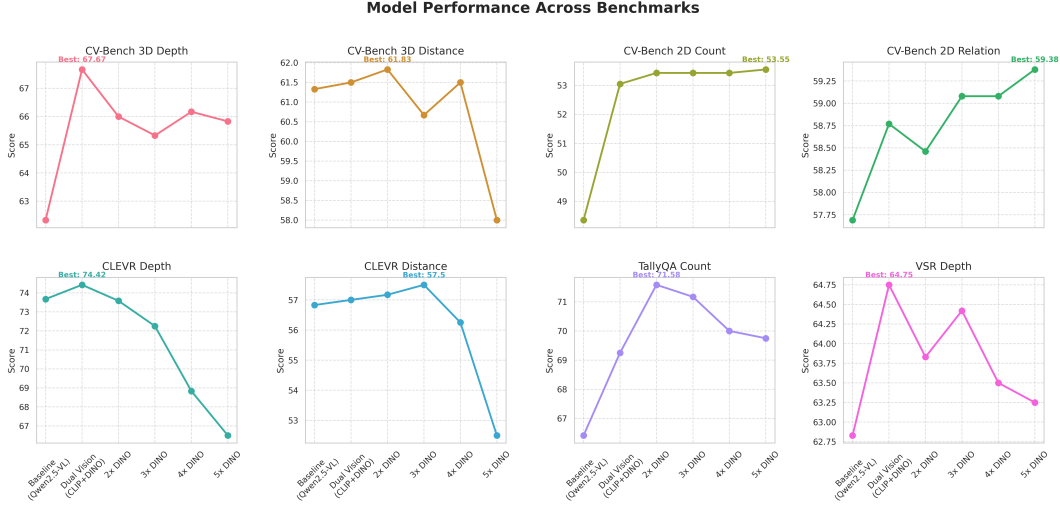


Figure 5: Ablation study evaluating model performance across varying fusion scaling factors (α). The graphs illustrate the performance trajectories across multiple spatial reasoning benchmarks (CV-Bench, CLEVR, TallyQA, VSR), demonstrating an optimal balance point between geometric precision and semantic comprehension before performance degrades.

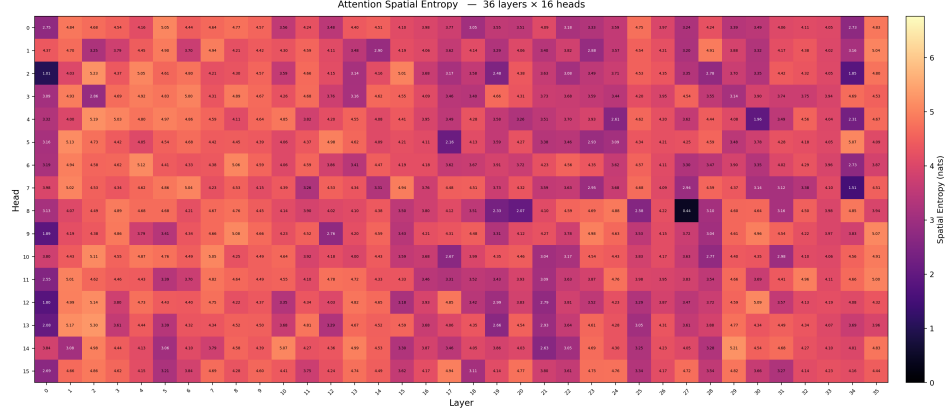
of the semantic alignment necessary to accurately interpret and resolve complex textual queries. This fundamental trade-off between geometric precision and semantic comprehension ultimately yields an inverted U-shaped performance curve, with optimal visual reasoning achieved at an intermediate value of α .

5 Conclusion

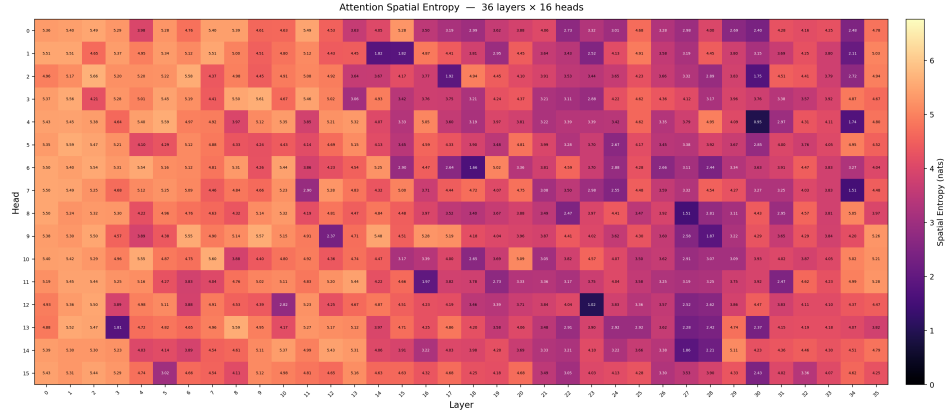
In this work, we addressed the fundamental limitations of contemporary VLMs in fine-grained spatial reasoning, a deficiency primarily stemming from the spatially homogeneous representations produced by standard CLIP-variant encoders. To overcome this representational bottleneck, we proposed a dual-encoder architecture that integrates the dense, spatially discriminative features of a self-supervised DINOv2 backbone into the Qwen2.5-VL baseline. By employing a straightforward token-level fusion mechanism, we successfully isolated and quantified the impact of supplementary geometric priors on the language model’s spatial acuity.

Our rigorous evaluation across spatially demanding benchmarks—including CV-Bench, CLEVR, TallyQA, and VSR—demonstrates that the augmented framework significantly outperforms the baseline. The integration of DINOv2 proved particularly highly effective in tasks requiring dense spatial awareness, such as object counting and depth estimation. Corroborating these quantitative gains, our qualitative analyses utilizing PCA and patch-level cosine similarity confirmed that the dual-encoder architecture successfully resolves spatial diffusion, facilitating sharply localized semantic boundaries and concentrated intra-object feature affinity.

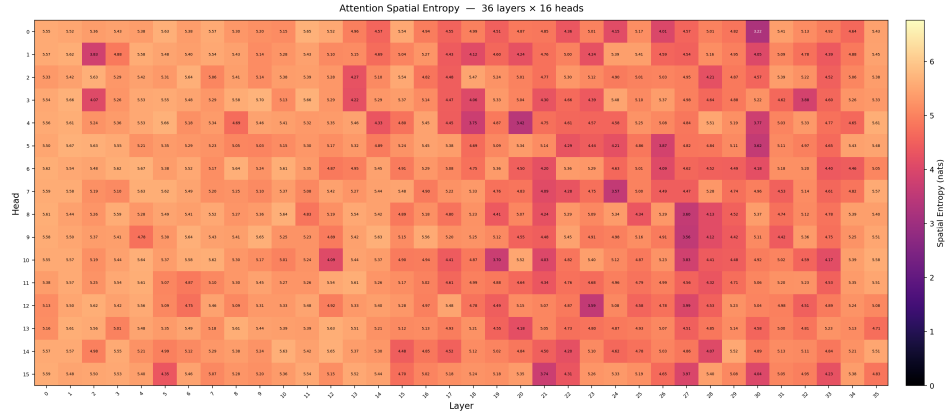
Furthermore, our ablation study on the fusion scaling factor α revealed a critical internal dynamic governing multimodal integration. Through the lens of layer-wise spatial entropy, we established that while injecting structural priors significantly sharpens the language model’s spatial attention and enhances visual grounding, an over-reliance on these explicitly geometric features disrupts the semantic alignment required to parse complex textual queries. This fundamental trade-off between spatial precision and semantic comprehension dictates that an optimal balance is required to maximize reasoning capabilities. Ultimately, this investigation validates the necessity and efficacy of explicitly spatially-aware representation streams, providing a robust architectural blueprint for the next generation of geometrically grounded Vision-Language Models.



(a) Baseline spatial entropy



(b) Spatial entropy at $\alpha = 8$



(c) Spatial entropy at high $\alpha = 15$

Figure 6: Layer-wise spatial attention entropy across varying fusion scaling factors (α). The heatmaps demonstrate that a moderate structural prior ($\alpha=8$) sharply focuses the language model’s spatial attention compared to the baseline. However, an excessively high scaling factor ($\alpha=15$) disrupts semantic alignment, significantly increasing entropy in deeper layers and correlating with a degradation in overall performance.

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